

Lesson 9 — Nested Loops & the Loop Variable

Matches LEGO Teacher Portal lesson: "Tick-Tick Trouble".

Objective (I can...): I can nest loops and use the loop counter to make patterns that change each pass.

CS concept: Nested iteration; using the loop index/variable · **Builds on:** Lesson 8 (loops and controlled stopping).

AI Connection (2 min): Training nests loops: for each epoch (outer), loop through every example (inner). *First-principles question:* "If it's 5 epochs over 100 examples, how many single updates run in total — and why does nesting make that easy to express?"

Materials: LEGO CS & AI kit + device per pair, grid paper, "broken program" cards, notebooks. **Model reuse:** Loops model from Lesson 6 — code only.

60-minute flow - Warm-up (8): "Rows and columns" — pairs compute total actions for several outer×inner combinations and spot the multiplication pattern. Connect to nested loops. - **Concept + demo (10):** Build an inner loop (tone 3×) inside an outer loop (repeat 4×); run it so students hear 4 sets of 3. Then show using the loop counter so each outer pass drives a little farther. (*Rotate format:* "trace it on paper first" — predict total and order.) - **Build/Code (30):** Pairs code a nested loop where the inner body repeats and the outer pass changes something using the loop index (e.g., drive distance grows each pass). **Debug task (5 min):** receive a program with inner/outer counts swapped; find why the total is wrong and fix it, logging the diagnosis. *Swap at 15 min.* - **Challenge (7):** Change the inner or outer count and predict the new total before running. - **Share + pack-up (5):** Pairs demo, state the total inner runs, and explain how the loop variable changed each pass. Model stays built.

Pair roles: Driver builds the inner loop, Navigator adds the outer and tracks the index. *Swap at 15 min. (Trio: third leads the debug.)*

Differentiation: *Support* — a worked nested example without the loop variable. *Extension / fast-finisher* — use the index in two places (distance and tone pitch). *Reteach* — build and test the inner loop alone first, then wrap the outer around it.

Exit check: Pair runs a working nested loop and can explain inner vs. outer counts and how the index was used. **At-home (optional):** Write a nested loop in pseudocode for a seating chart (rows of seats); state the total seats. ~10 min.